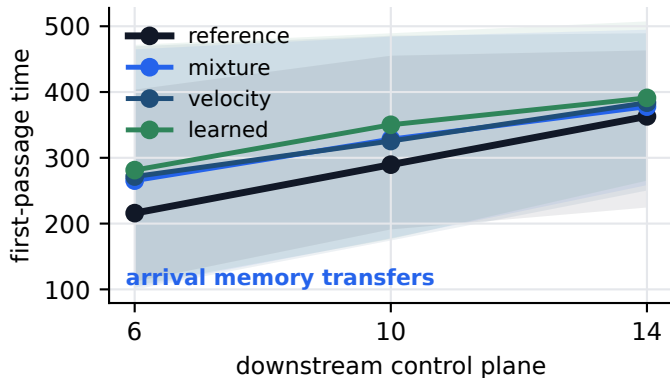


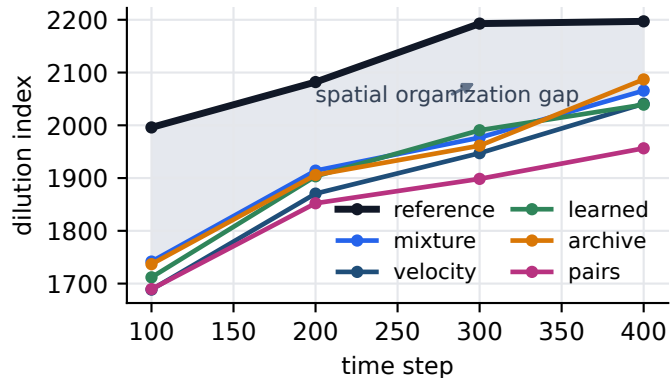
A model can arrive correctly and still forget the plume

Breakthrough-only validation selects arrival memory, but other observables expose unsafe forgetting.

a arrival preserved



b dilution forgotten



c Breakthrough-only selection favors velocity memory, not universal adequacy

